



哈爾濱工業大學(深圳)
HARBIN INSTITUTE OF TECHNOLOGY, SHENZHEN

高级语言程序设计 High-level Language Programming

Lecture 1 Introduction

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School of Information Science and Technology

Let's Get Connected!

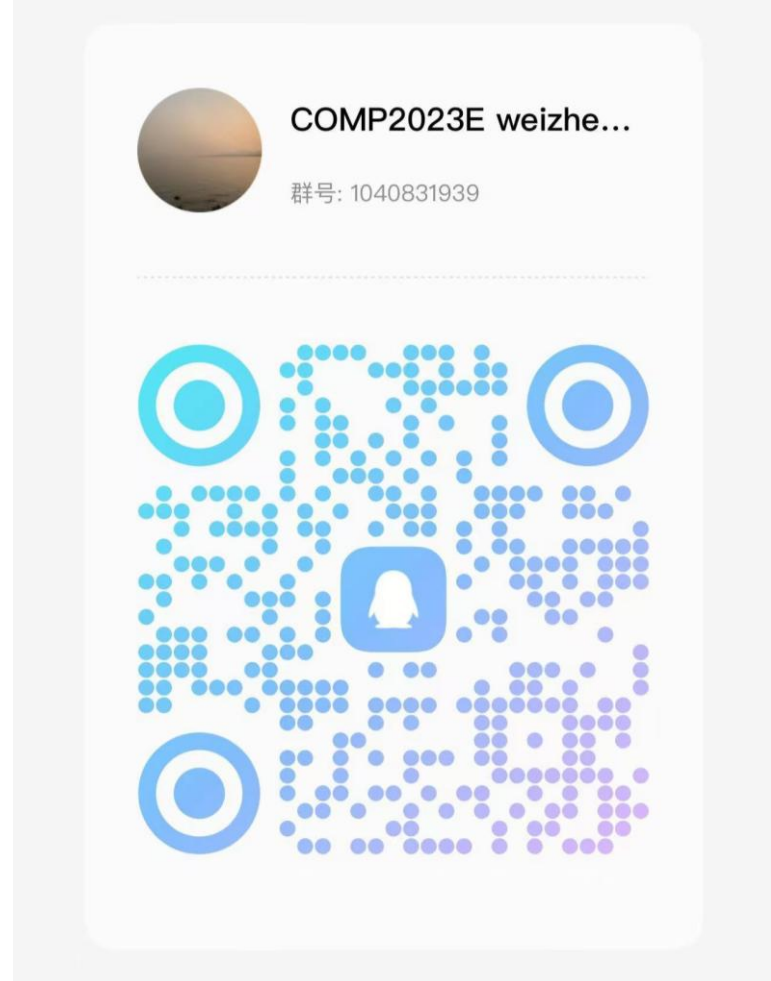
- Course related discussion on QQ
 - Posting homework
 - Sharing course materials
 - Course related Q&A
- **Step 1:** Download and install QQ on your mobile/PC: im.qq.com/index



 iOS



 Android



- **Step 2:** Inside your QQ, scan the QR code above
- Group number **1040831939**

Personal Information

Prof. Dr. **Weizheng Zhang** (Chinese name: 张伟铮)

- School of Information Science and Technology
- Research interests: **space information networks, mmWave communications, and massive MIMO.**



Academic Background

- Postdoctoral Researcher, University of New South Wales, Sydney, Australia (2020-2023)
- Ph.D., Wireless Communications, School of Electrical Engineering and Telecommunications, University of New South Wales. (2015-2019)

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Office room: L1225

Teaching Assistant: Aibiao Zhang, Chengyao Shi and Yixin Zhang

Course Objectives

- Master the fundamental approaches to **solve problems with computers**
- Master the basic knowledge of **high-level programming language** (C++)
- Master the fundamental methods of program design and implementation
- Acquire the ability to solve practical problems using computers
- Acquire the basic skills for programming and debugging

Assessment & Grades

- 6 weeks of Lectures for the Computer Science (**CS**) part
 - Monday 2pm-3:45pm, T2-806, week 1-5, week 7
 - Wednesday 10:30am-12:15pm, T2-806, week 1-4, week 7
 - Sunday 10:30am-12:15pm, T2-806, week 4 only (Sep 28)
- 4 times of Labs
 - Tuesday, week 2, 2pm-3:45pm, T2-617
 - Monday, week 5, 7, 9, 10:30am-12:15pm, T2-617
- Course assignments and exams
 - **Homework** (2% each) x 10 = **20% points**
 - **Laboratory Assignments** (10% each) x 4 = **40% points**
 - **Final Examination** x 1 = **40% points**
- Homework & Lab submission and grading
 - Teaching Assistant: **Aibiao Zhang, Chengyao Shi and Yixin Zhang**

Class Attendance

- Class attendance is required and will be checked randomly.

If you miss over 1/3 class (miss > 4 lectures), you will be disqualified to attend the Final Examination (Loss of 40% points)

- **Recommended:** Bring your laptop to class, for in-class exercises and try it yourself examples.

Expectations

- Attend lectures on time
 - Complete assignments on your own and submit on time
 - Engage in class discussion
 - Communicate with instructor if need help
-
- All lectures are given in **English**
 - All assignments and exams must be completed in **English**

Textbook

Paul Kelly, Xiaohong Su, *Learning C++ through English and Chinese (2nd Edition)*, Publishing House of Electronics Industry, 2016



Other online references and materials

- Stanley B. Lippman, Josée Lajoie, Barbara E. Moo, [*C++ Primer \(Fifth Edition\)*](#), Addison-Wesley, 2013
- Mark Allen Weiss, [*Data Structures and Algorithm Analysis in C++ \(Fourth Edition\)*](#), Pearson Education, 2014

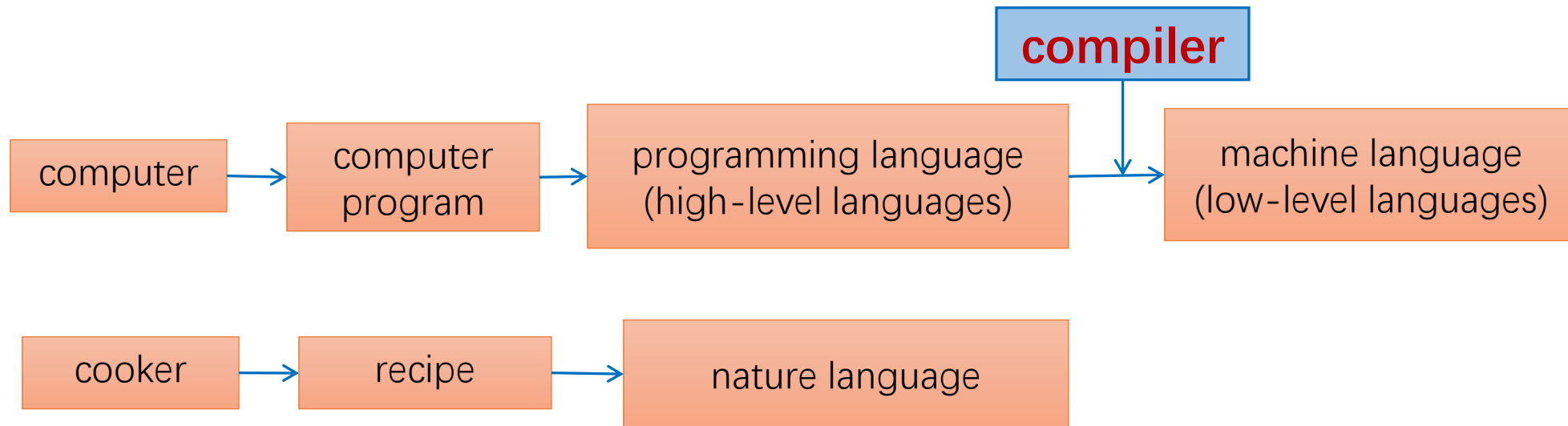
Introduction

Course Overview

- 1.1 What is a computer program?
- 1.2 How to develop a computer program?
- 1.3 The brief history of the C++ language
- 1.4 The common basic standards in the C++
- 1.5 How to use the online programming tool, e.g. "onlinegdb"
- 1.6 How to use the C++ programming tool
- 1.7 How to learn C++

1.1 What is a computer program?

- **Computer program** : A computer performs all of these tasks by following a predefined set of instructions.



1.1 What is a computer program?

- **Programming language:**

```
#include <iostream>

int main()
{
    std::cout << "I love c++ programing" << std::endl;

    return 0;
}
```

- **Natural language:**

Please print “I love c++
programing” onto the screen.

1.2 Developing a computer program

- Define and understand the problem to be solved - - - - analysis phase
- How is it to be done? - - - - - design phase
- Writing, compiling and testing C++ programs

1.2 Developing a computer program

Program development cycle

- Step 1: Design the program.
 - Each program has to be individually designed to implement the overall solution developed at the analysis and design phases.
 - Check its logic before starting to write the program
- Step 2: Write the program.
 - The C++ program instructions are typed into a file using a *text editor*



source code or the program code



source file

1.2 Developing a computer program

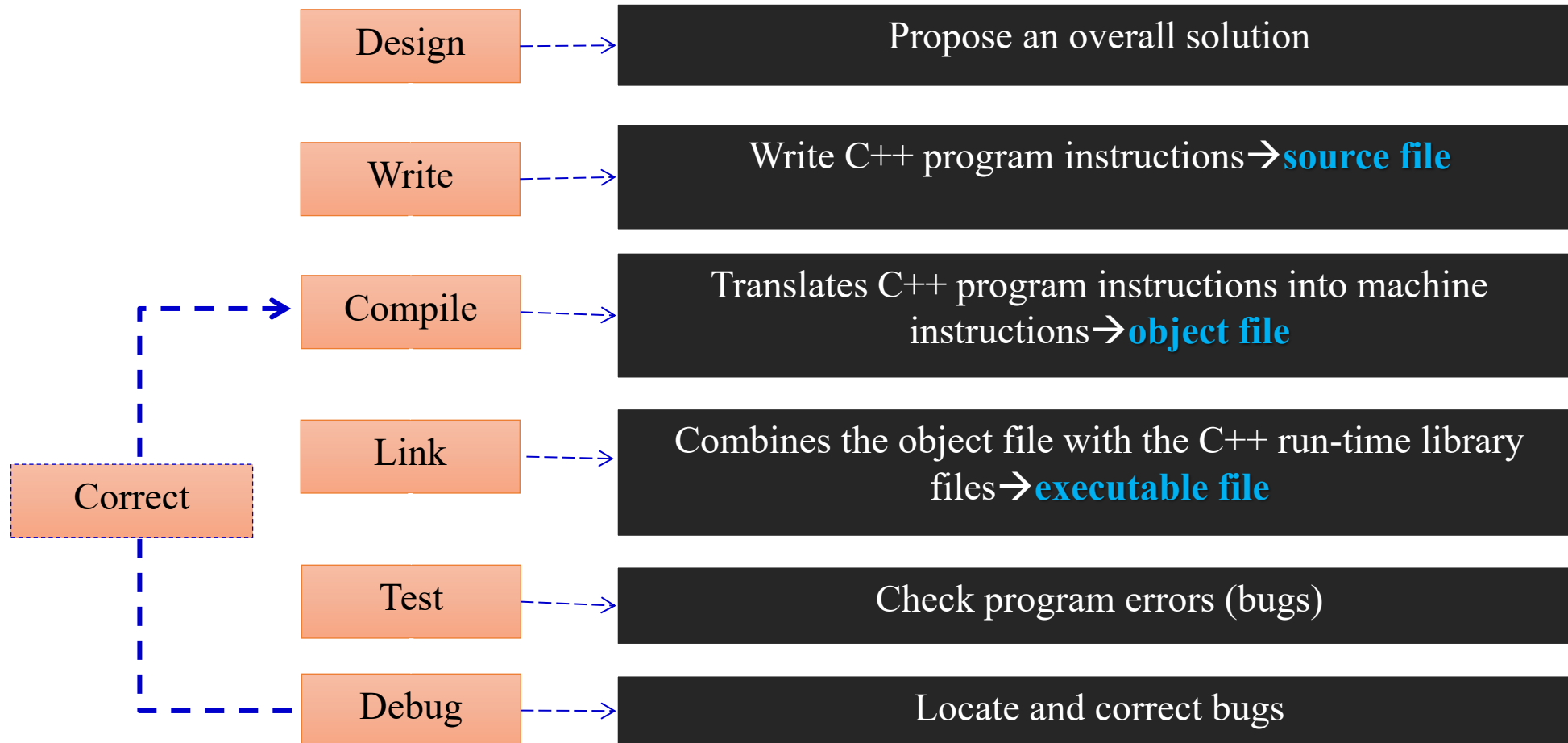
- Step 3: Compile the program.
 - Translate the C++ program instructions to machine instructions through a compiler-- *object code*, and stores the object code in an *object file*.
 - Errors ---- *compile-time syntax error*: missing punctuation , misspellings,...
 - Warning message: not as serious as a syntax error, not prevent the program from being compiled, may be a problem and should be investigated
- Step 4: Link the program.
 - Combining the object file of the program with other object files from the C++ run-time library to form an *executable file*

1.2 Developing a computer program

- Step 5: Test the program.
 - *run-time error*—causes the program to stop before it has completed its task
 - *logic errors(bugs)*---The program may complete its task but produce incorrect results
- Step 6: Debug the program.
 - The process of locating and correcting program errors is called *debugging*
 - Many compilers have tools that can be used to help locate bugs
 - Correcting bugs involves going back to step 2, try to catch errors as early as possible

1.2 Developing a computer program

Program development cycle



1.3 Brief history of C++

- 1972, AT&T, Bell Lab. Dennis Ritchie, C programming language
- 1980, Bell Lab. Bjarne Stroustrup, “C with Classes”
- 1983, C++ was developed by Bjarne Stroustrup at AT&T Bell Laboratories
- 1985, the first commercial version of C++
- 1997, ANSI (American National Standards Institute) C++ (standard C++)

C++ is always evolving !!!

1.4 Common standards in the C++

- The example programs used in the text book conform to the ANSI/ISO C++ standard.
 - Not all compilers conform to this standard, so some compilers may not correctly compile the example programs
 - Some of the example programs may have to be modified for use with these other compilers
 - See the website for more details

1.5 Online Programming

onlinegdb: <https://www.onlinegdb.com/>



The screenshot displays the OnlineGDB web interface. On the left is a blue sidebar with the OnlineGDB logo and navigation links: IDE, My Projects, Classroom (marked 'new'), Learn Programming, Programming Questions, Sign Up, and Login. The main area features a toolbar with icons for file operations, Run, Debug, Stop, Share, Save, and Beautify. Below the toolbar, the file 'main.cpp' is open, showing a C++ program. The code includes a multi-line comment, an `#include <iostream>` directive, and a `main` function that prints 'Hello World' and returns 0. The language is set to C++ in the top right. At the bottom, there is an 'input' field and a 'Command line arguments:' input box.

```
1  /**
2
3  Welcome to GDB Online.
4  GDB online is an online compiler and debugger tool for C, C++, Python, Java, PHP, Ruby, Perl,
5  C#, OCaml, VB, Swift, Pascal, Fortran, Haskell, Objective-C, Assembly, HTML, CSS, JS, SQLite, ProLog.
6  Code, Compile, Run and Debug online from anywhere in world.
7
8  *****/
9  #include <iostream>
10
11 int main()
12 {
13     std::cout<<"Hello World";
14
15     return 0;
16 }
```

1.6 Offline Programming

CodeBlocks

download [CodeBlocks](#) and install it



Microsoft Windows

File	Download from
codeblocks-20.03-setup.exe	FossHUB or Sourceforge.net
codeblocks-20.03-setup-nonadmin.exe	FossHUB or Sourceforge.net
codeblocks-20.03-nosetup.zip	FossHUB or Sourceforge.net
codeblocks-20.03mingw-setup.exe	FossHUB or Sourceforge.net
codeblocks-20.03mingw-nosetup.zip	FossHUB or Sourceforge.net
codeblocks-20.03-32bit-setup.exe	FossHUB or Sourceforge.net
codeblocks-20.03-32bit-setup-nonadmin.exe	FossHUB or Sourceforge.net
codeblocks-20.03-32bit-nosetup.zip	FossHUB or Sourceforge.net
codeblocks-20.03mingw-32bit-setup.exe	FossHUB or Sourceforge.net
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1.6 Offline Programming

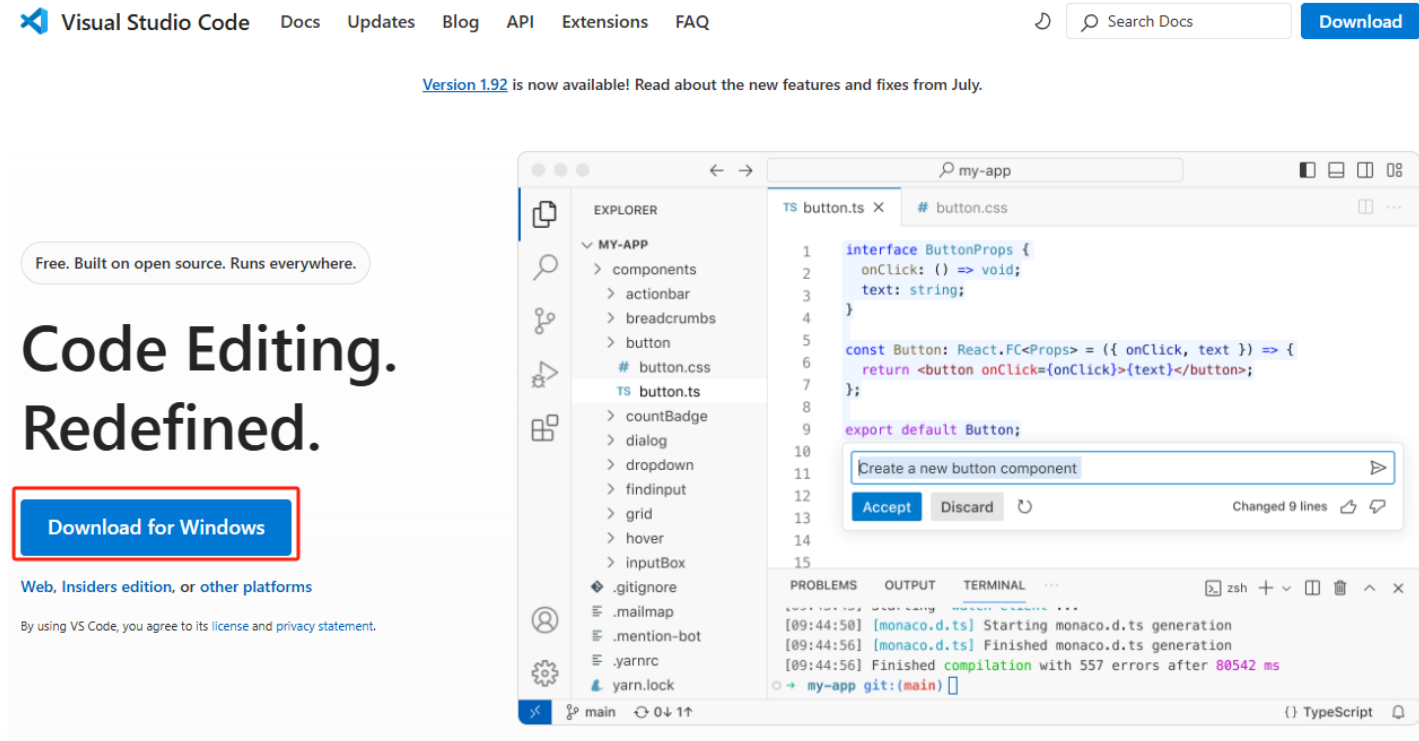
CodeBlocks

The installation package includes the compilation environment, and the installation will automatically find the environment path. Therefore, you can directly create new files to write and run C++ programs.

1.6 Offline Programming

Visual Studio Code

- 1. download [Visual Studio Code](#) and install it



1.6 Offline Programming

- 2. download [MinGW-W64](#) and install it

▼ Assets 15

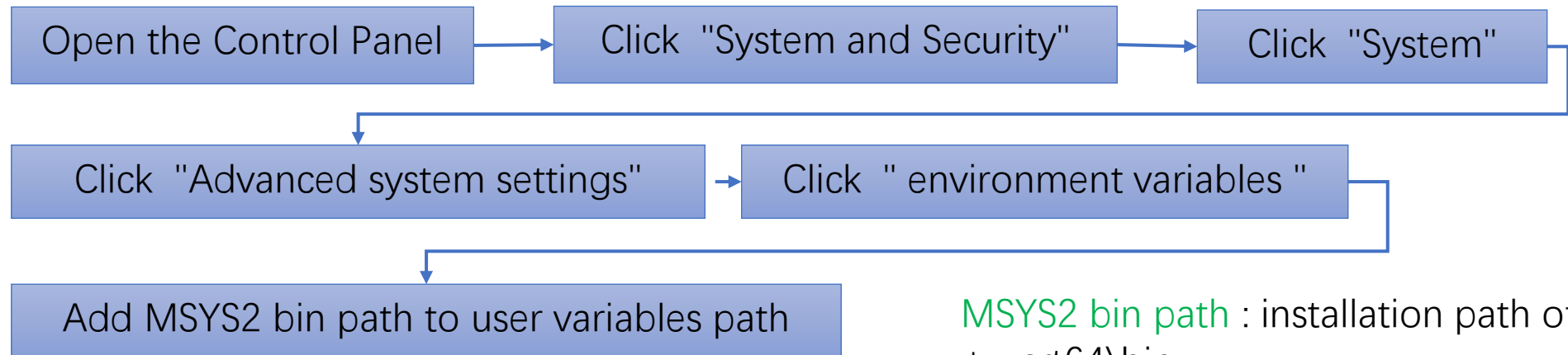
 msys2-base-x86_64-20240727.packages.txt	1.46 KB	3 weeks ago
 msys2-base-x86_64-20240727.sfx.exe	46.5 MB	3 weeks ago
 msys2-base-x86_64-20240727.sfx.exe.sha256	101 Bytes	3 weeks ago
 msys2-base-x86_64-20240727.sfx.exe.sig	566 Bytes	3 weeks ago
 msys2-base-x86_64-20240727.tar.xz	46.7 MB	3 weeks ago
 msys2-base-x86_64-20240727.tar.xz.sha256	100 Bytes	3 weeks ago
 msys2-base-x86_64-20240727.tar.xz.sig	566 Bytes	3 weeks ago
 msys2-base-x86_64-20240727.tar.zst	47.5 MB	3 weeks ago
 msys2-base-x86_64-20240727.tar.zst.sha256	101 Bytes	3 weeks ago
 msys2-base-x86_64-20240727.tar.zst.sig	566 Bytes	3 weeks ago
 msys2-x86_64-20240727.exe	79.4 MB	3 weeks ago
 msys2-x86_64-20240727.exe.sha256	92 Bytes	3 weeks ago
 msys2-x86_64-20240727.exe.sig	566 Bytes	3 weeks ago
 Source code (zip)		Apr 5
 Source code (tar.gz)		Apr 5

1.6 Offline Programming

- 3. After finishing the installation, open MSYS2 and enter the following command to install the toolchain:

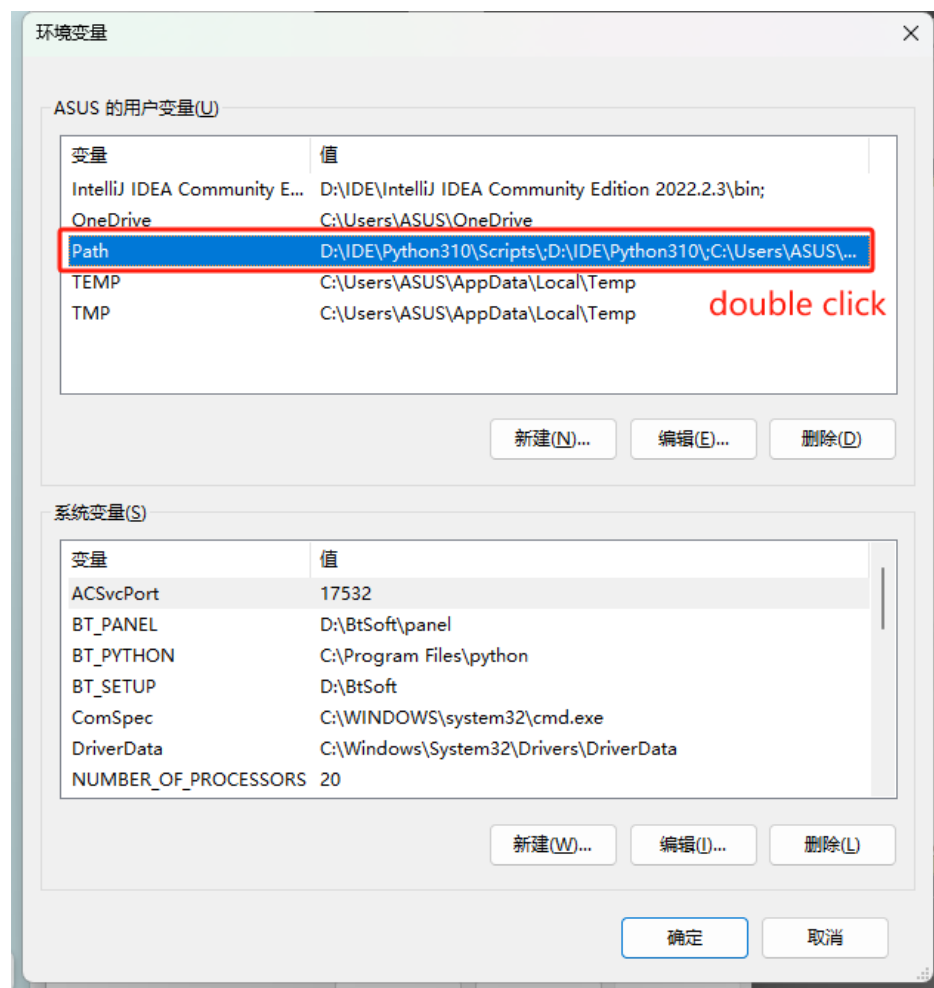
```
pacman -S --needed base-devel mingw-w64-ucrt-x86_64-toolchain
```

- 4. Configure system environment variables



MSYS2 bin path : installation path of MSYS2
+ ucrt64\bin
for example, C:\msys64\ucrt64\bin

1.6 Offline Programming



1.6 Offline Programming

- 5. use the following commands to test the installation of environment:

```
C:\Windows\system32\cmd.e: x + v
Microsoft Windows [版本 10.0.22631.3296]
(c) Microsoft Corporation. 保留所有权利。

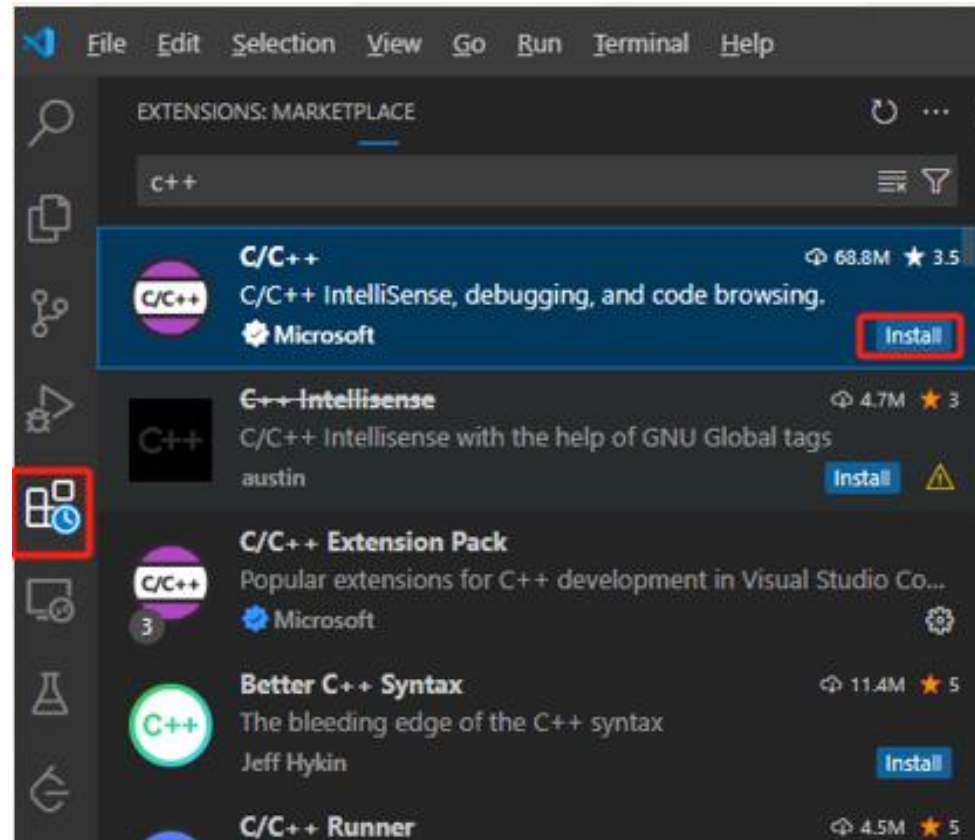
C:\Users\27437>gcc --version
gcc (Rev3, Built by MSYS2 project) 13.2.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

C:\Users\27437>g++ --version
g++ (Rev3, Built by MSYS2 project) 13.2.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

C:\Users\27437>gdb --version
GNU gdb (GDB) 14.1
Copyright (C) 2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
```

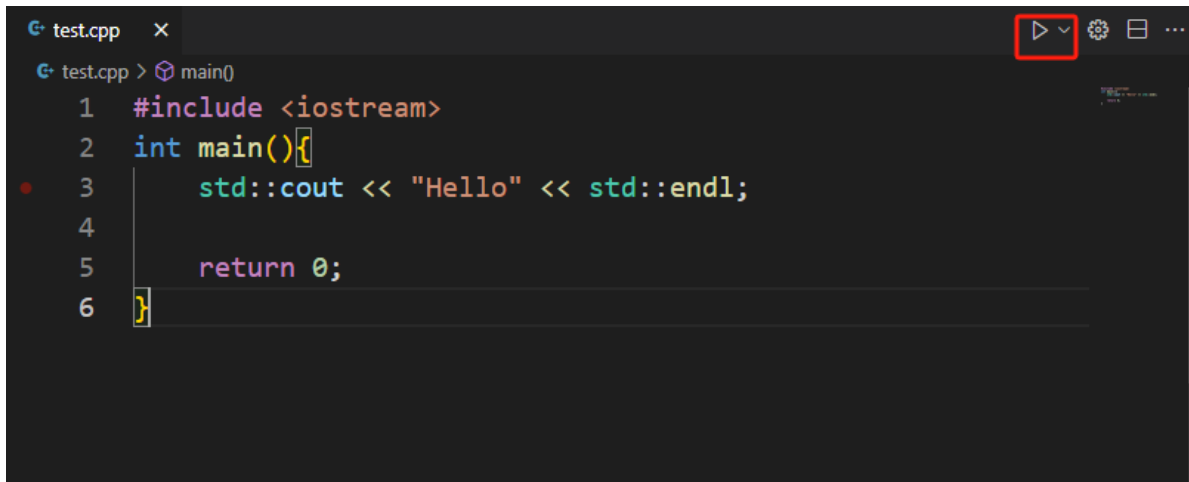
1.6 Offline Programming

- 6. open Visual Studio Code, search for "C++" in extensions and install it:



1.6 Offline Programming

- 7. test whether vscode can run c++ files



```
test.cpp x
test.cpp > main()
1  #include <iostream>
2  int main(){
3      std::cout << "Hello" << std::endl;
4
5      return 0;
6  }
```

or use following commands:

```
g++ -o test test.cpp
./test
```

1.7 Learning C++

- A successful approach to learning to program in C++ depends on large amounts of **practice**.
 - exercises at the end of each chapter.
 - solutions to selected exercises are at the web site for this book
www.hxedu.com.cn
- Other learning materials

[Learn C++ – Skill up with our free tutorials \(learncpp.com\)](http://learncpp.com)

https://www.bilibili.com/video/BV1oD4y1h7S3/?spm_id_from=333.337.search-card.all.click&vd_source=1e2cfa9e6c262e6d189fe4a48f2c1aac

[Learn C++ | Codecademy](https://www.codecademy.com/learn/cplusplus)

Homework 1

- 1. How to develop a C++ program?
- 2. Use CodeBlocks, Visual Studio Code or online programming to write a C++ program:

Display the following two sentences on the screen:

Hello C++.

I like programming.

The output is formatted as two lines.